Live Students Room System (Next.js + Node.js) – Architecture Guide

Overview

This document describes how to build a live classroom system where students can join rooms, collaborate using a shared whiteboard, participate in voice/video discussions, share screens, and allow mentors to manage the session.

Core Technologies

Frontend: Next.js (React framework)

Backend: Node.js with Express.js

Database: MongoDB

Real■time Communication: Socket.IO (WebSockets)

Video / Voice / Screen Sharing: WebRTC

Main Features

1. Room Join System – Students and mentors join a room using a unique room ID.
2. Shared Whiteboard – Collaborative drawing using HTML Canvas with real■time sync via Socket.IO.
3. Voice and Video Discussion – WebRTC peer connections for live communication.
4. Screen Sharing – Browser screen sharing using getDisplayMedia().
5. Mentor Controls – Mentor can manage the room, share screen, and guide students.

System Flow

User Login → Create/Join Room → WebSocket Connection → Users Join Room → WebRTC Connections Established → Real■time Collaboration Begins.

Database Structure (MongoDB)

Users Collection: id, name, email, role (student/mentor)

Rooms Collection: roomId, mentorId, students[], createdAt

Whiteboard Data: roomId, drawingData[]

Useful Libraries

socket.io

simple-peer (WebRTC abstraction)

fabric.js (advanced whiteboard features)

uuid (unique room IDs)

Production Enhancements

Use TURN/STUN servers for reliable WebRTC connections.

Use Redis for scaling socket connections.

Deploy backend on cloud infrastructure such as AWS, GCP, or DigitalOcean.

Technology Stack Summary

Layer	**Technology**
Frontend	Next.js, React, HTML Canvas
Backend	Node.js, Express.js
Realtime	Socket.IO
Video/Audio	WebRTC
Database	MongoDB